

FGFR2 fusion-hotspot benchmark report

Study: brca_tcga_pan_can_atlas_2018

Abstract

This report analyzes FGFR2 gene fusions from the brca_tcga_pan_can_atlas_2018 study, covering 5 fusion events. 4/5 fusions (80.0%) were in-frame. 2/5 fusions (40.0%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test ($p=0.6$, not statistically significant at $\alpha=0.05$). No literature benchmark is configured for FGFR2.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 5 analyzed fusion events, identifying 5 distinct fusion partner genes. The most frequent partner was AP1M1, observed in 1 event.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 2/4 (50.0%) of in-frame fusions retained the domain, $p=0.6$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.020979.

Domain disruption

Domain-disruption analysis was skipped: Algorithm failed: ValueError: no in-frame protein-fusion events have a known domain-retention state

Cutpoint detection

Cutpoint detection scanned 5 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 621 aa (exon 13; permutation-corrected $p=0.200799$, not statistically significant at $\alpha=0.05$). This is -136 aa from the nearest configured domain boundary at 757 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=2$) and "not_retained" ($n=3$). For Frame_status, retained: 2/2 (100.0%, 95% CI 34.2%-100.0%); not_retained: 2/2 (100.0%, 95% CI 34.2%-100.0%). Welch's t-test comparing Tumor_variant_count between groups gave means of unavailable vs unavailable, $p=$ unavailable.

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association produced results for this run; see the accompanying data for details.

Genomic position recurrence

Genomic position recurrence produced results for this run (4 row(s) in genomic_bin_recurrence, 1 row(s) in exact_position_recurrence, 1 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: FGFR2 has no gene_pair configured; joint-partner analysis was skipped.

Mutation cooccurrence

Mutation cooccurrence reported: No cohort samples carried the configured comparator alteration (FGFR2 point_mutation (S252W)); its co-occurrence statistics reflect zero comparator-positive samples.

Window detection

Window detection reported: no candidate window of the configured widths (25, 50, 100, 200) met the minimum-event-count guard (min_events_per_window=4) on both sides of the split

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
AP1M1	1	0.2	0.16782147827453991	None	1.0	0.342380227506653	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.35774174039912526	1
CTNNA3	1	0.2	0.16782147827453991	None	0.2731958762886598	0.342380227506653	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.22146596720324907	2
CASP7	1	0.2	0.16782147827453991	None	0.24742268041237114	0.342380227506653	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.2166334929764449	3
CCDC6	1	0.2	0.16782147827453991	None	0.24742268041237114	0.342380227506653	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.2166334929764449	4
ENPP2	1	0.2	0.16782147827453991	None	0.0	0.342380227506653	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.17024174039912535	5

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
427	0	2	1	2	0.0	1.0	-0.0
480	0	2	2	1	0.0	0.4000000000000001	0.39794000867203755
621	0	2	3	0	0.0	0.10000000000000002	0.9999999999999999

domain_retention tables

frame_domain_contingency_table

2	0
2	1

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	5	2	1	2	0.16967509025270758

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
11	3	5	0.6

frequency tables

Partner_gene_counts

Partner_gene	Event_count
AP1M1	1
CASP7	1
CCDC6	1
CTNNA3	1
ENPP2	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
10	121483000	121484000	2	EVT-TCGA-AN-A0AL-01-2, EVT-TCGA-D8-A13Z-01-3
10	121496000	121497000	1	EVT-TCGA-AQ-A04L-01-1
10	121503000	121504000	1	EVT-TCGA-AO-A03V-01-4
10	121515000	121516000	1	EVT-TCGA-LD-A7W5-01-5

exact_position_recurrence

chromosome	position	n_events	event_ids
10	121483698	2	EVT-TCGA-AN-A0AL-01-2, EVT-TCGA-D8-A13Z-01-3

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
767	2	1	0	False

mutation_cooccurrence tables

cooccurrence_results

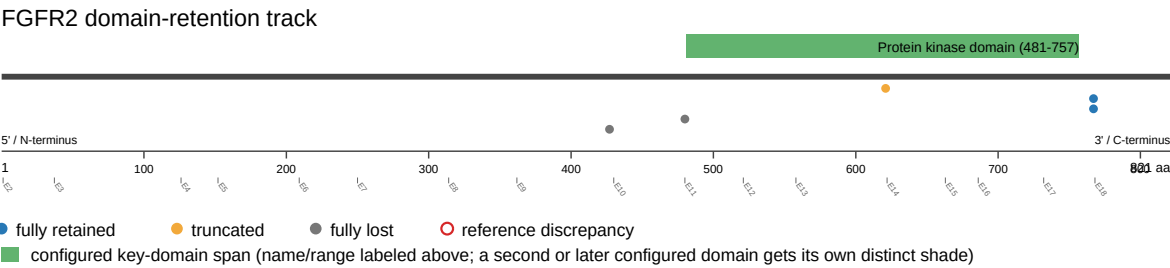
comparator_gene	alteration_type	protein_change	label	contingency_table	odds_ratio	p_value	direction	cohort_sample_count	fusion_positive_sample_count	comparator_altered_sample_count	both_altered_sample_count
FGFR2	point_mutation	S252W	FGFR2 point_mutation (S252W)	[0, 5], [0, 1079]	None	1.0	indeterminate	1084	5	0	0

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-AQ-A04L-01-1	TCGA-AQ-A04L-01		Breast Cancer	IDC	FGFR2-AP1M1	AP1M1	in-frame	five_prime	121496532	13	621	False	621	True	exact match	disrupted	0.5090252707581228	True			Protein kinase domain	NA FGFR2-AP1M1 Fusion	in-frame
EVT-TCGA-AN-A0AL-01-2	TCGA-AN-A0AL-01		Breast Cancer	IDC	FGFR2-CASP7	CASP7	in-frame	five_prime	121483698	17	767	False	767	True	exact match	retained	1.0	False	Protein kinase domain			NA FGFR2-CASP7 Fusion	in-frame
EVT-TCGA-D8-A13Z-01-3	TCGA-D8-A13Z-01		Breast Cancer	IDC	FGFR2-CCDC6	CCDC6	in-frame	five_prime	121483698	17	767	False	767	True	exact match	retained	1.0	False	Protein kinase domain			NA FGFR2-CCDC6 Fusion	in-frame
EVT-TCGA-AO-A03V-01-4	TCGA-AO-A03V-01		Breast Cancer	IDC	FGFR2-CTNNA3	CTNNA3	unknown	five_prime	121503790	10	480	False	479	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		Protein kinase domain		NA FGFR2-CTNNA3 Fusion	frameshift
EVT-TCGA-LD-A7W5-01-5	TCGA-LD-A7W5-01		Breast Cancer	IDC	FGFR2-ENPP2	ENPP2	in-frame	five_prime	121515123	9	427	False	427	True	exact match	lost	0.0	False		Protein kinase domain		NA FGFR2-ENPP2 Fusion	in-frame

Figures

domain retention outliers



fusion schematic

